

DATA 430 Technical Report Assignment 2: Bayesian Classification	Matthew Kolonich
Stroke Prediction Using Naïve Bayes Classification	
URL to dataset: Stroke Prediction Dataset	

This template should be used in conjunction with the assignment instructions. The size of the text area below will expand to the length of your response; the area should not be interpreted as a required or suggested length of response. Responses within the text area should be single spaced with Times New Roman 12pt font. The body of the document will likely be 6-9 pages, not including the Appendix; length may vary depending on specifics of the analysis and the dataset. As needed, APA format in-text citations should be included, along with a full references list at the end of the document.

Overview
Problem Domain: give some background and context about the problem domain (application area). For instance, if you are doing the analysis for predicting heart disease, provide some context about the disease and include some interesting statistics about it. Also, discuss how the method is relevant for the chosen problem. Stroke is one of the leading causes of death and long-term disability worldwide. Healthcare organizations continually seek ways to identify patients who may be at a higher risk of experiencing a stroke. This is so preventative interventions can be implemented before a life-threatening event occurs. Advances in machine learning have created new opportunities to analyze large healthcare datasets and identify patterns that may not be immediately obvious through traditional analysis alone. As healthcare systems continue to collect large amounts of patient information, machine learning techniques can assist medical professionals in making more informed decisions. This can be great for patient risk factors and treatment planning. The problem domain for this study focuses on predicting the likelihood of a patient experiencing a stroke. This is based on demographic, lifestyle, and medical characteristics. The dataset includes variables such as age, hypertension status, heart disease history, average glucose level, body mass index (BMI), smoking status, marital status, residence type, and employment type. These variables provide information that may contribute to an individual's overall stroke risk profile. I find this problem domain important because stroke prediction represents a real-world application. Machine learning may have a meaningful impact on people's lives. Unlike some business applications that focus mainly on profit or efficiency, healthcare prediction models can help identify individuals who may benefit from earlier medical intervention. A timely diagnosis or preventative treatment could reduce the severity of a stroke or potentially prevent one from occurring altogether. This study uses Bayesian Classification. It specifically uses the Naïve Bayes algorithm, to examine relationships between patient characteristics and stroke outcomes. Bayesian

methods are useful in machine learning. They apply probability theory to classification problems. In this assignment, the model uses patient features as evidence and estimates the probability that a patient belongs to either the stroke or non-stroke class. The course lecture explains that Naïve Bayes is a supervised machine learning algorithm that applies Bayes' theorem while assuming that input features are conditionally independent given the target class (Bati, n.d.).

The importance of predictive modeling in healthcare is also supported by outside academic research. Beam and Kohane (2018) explain that machine learning is increasingly important in healthcare because large medical datasets can be analyzed to support clinical decision-making. This project examines how Bayesian Classification can be applied to stroke prediction using patient health data. The primary goal is not simply to produce a prediction, but to better understand whether a probability-based classification method can support earlier identification of stroke risk.

Objective: clearly state the objective of the analysis in relation to the kind of algorithm you are employing. Use specific language as to what question(s) you are trying to answer using the specific analysis/modeling type.

The objective of this study is to develop and evaluate a Bayesian Classification model, using the Gaussian Naïve Bayes algorithm, to predict whether a patient experienced a stroke. The dependent variable in this study is stroke. This is a binary classification variable where 0 represents no stroke and 1 represents stroke occurrence. The independent variables include age, hypertension, heart disease, average glucose level, BMI, gender, marital status, work type, residence type, and smoking status.

The Gaussian Naïve Bayes algorithm was selected because the dataset contains several continuous variables. These include age, average glucose level, and BMI. Gaussian Naïve Bayes assumes that numerical variables follow a normal distribution within each class. It also applies Bayes' Theorem to estimate the probability that an observation belongs to a particular category. According to Bati (n.d.), Bayesian Classification is a supervised learning method that uses probability theory to classify observations based on available evidence.

Another objective of this study is to evaluate how well a Bayesian Classification model performs on a healthcare dataset containing a highly imbalanced target variable. Only a small percentage of patients in the dataset experienced a stroke. This creates a challenging classification problem. Because of this imbalance, traditional measures such as overall accuracy are not sufficient by themselves. Additional metrics are needed. These include

precision, recall, F1-score, confusion matrix analysis, and ROC-AUC. They are required to fully evaluate model performance.

Personally, I was interested in determining whether Naïve Bayes could successfully identify high-risk patients. In healthcare applications, missing a patient who may be at risk for a serious condition can have significant consequences. Therefore, one of the primary goals of this project is to evaluate the model's ability to identify stroke cases while understanding the tradeoffs that occur between sensitivity and overall classification accuracy.

Analysis

Exploratory Analysis: describe the data including the source, the collection method, and variables. Perform exploratory analysis. Also, select few key variables (including the target variable for supervised learning) and study their distributions using plots such as histograms, box plot, bar chart, etc.

The Stroke Prediction Dataset contains 5,110 patient records and 12 variables. The dataset includes demographic, lifestyle, and medical information that may influence stroke risk. The variables include age, gender, hypertension status, heart disease history, average glucose level, BMI, smoking status, marital status, work type, residence type, and stroke outcome.

The exploratory analysis began by examining the dataset structure and identifying missing values. The dataset contained 201 missing BMI values. All other variables were complete. Because BMI may be an important predictor of stroke risk, these missing values were replaced using the mean BMI value. This approach preserved all available observations and avoided removing potentially useful information.

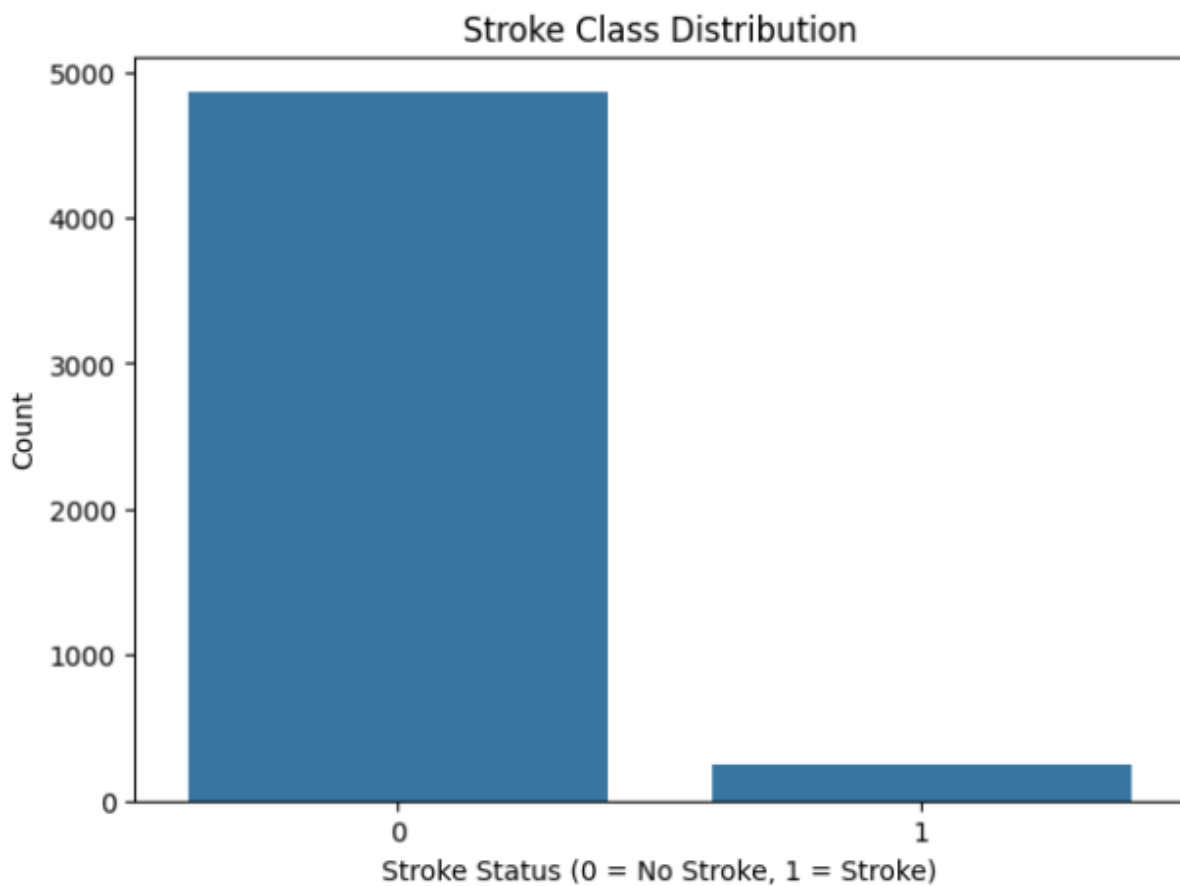
Descriptive statistics indicated that the average patient age was approximately 43.23 years. The average BMI was approximately 28.89. The average glucose level was approximately 106.15. The stroke variable had a mean value of approximately 0.049, indicating that only about 4.9% of patients in the dataset experienced a stroke.

The age distribution histogram demonstrated a distribution of patients across multiple age groups. The BMI distribution histogram showed that many patients fell within the overweight

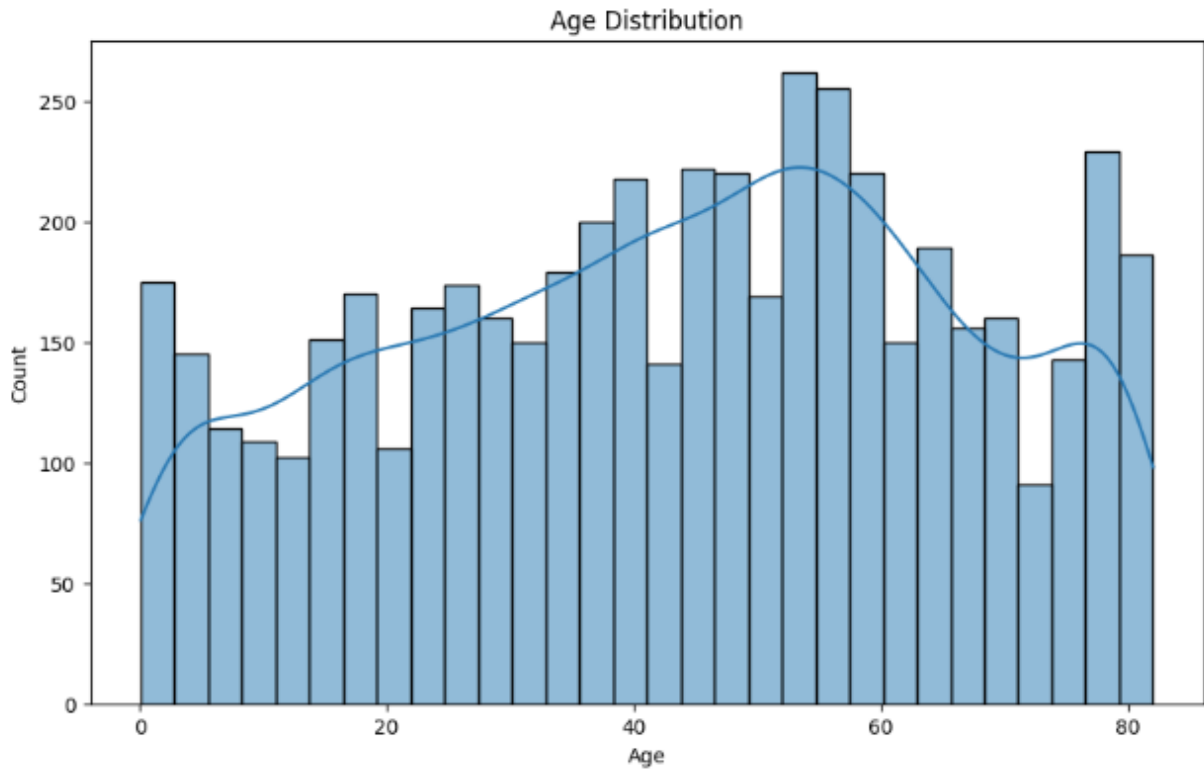
BMI range, with several high outliers present. The stroke distribution chart revealed a significant class imbalance, with substantially more non-stroke cases than stroke cases.

Kubat (2017) explains that highly imbalanced datasets can create challenges for classification algorithms because models may achieve misleading performance by favoring the majority class. This observation became important throughout the remainder of the analysis because stroke cases represented only a small fraction of the dataset.

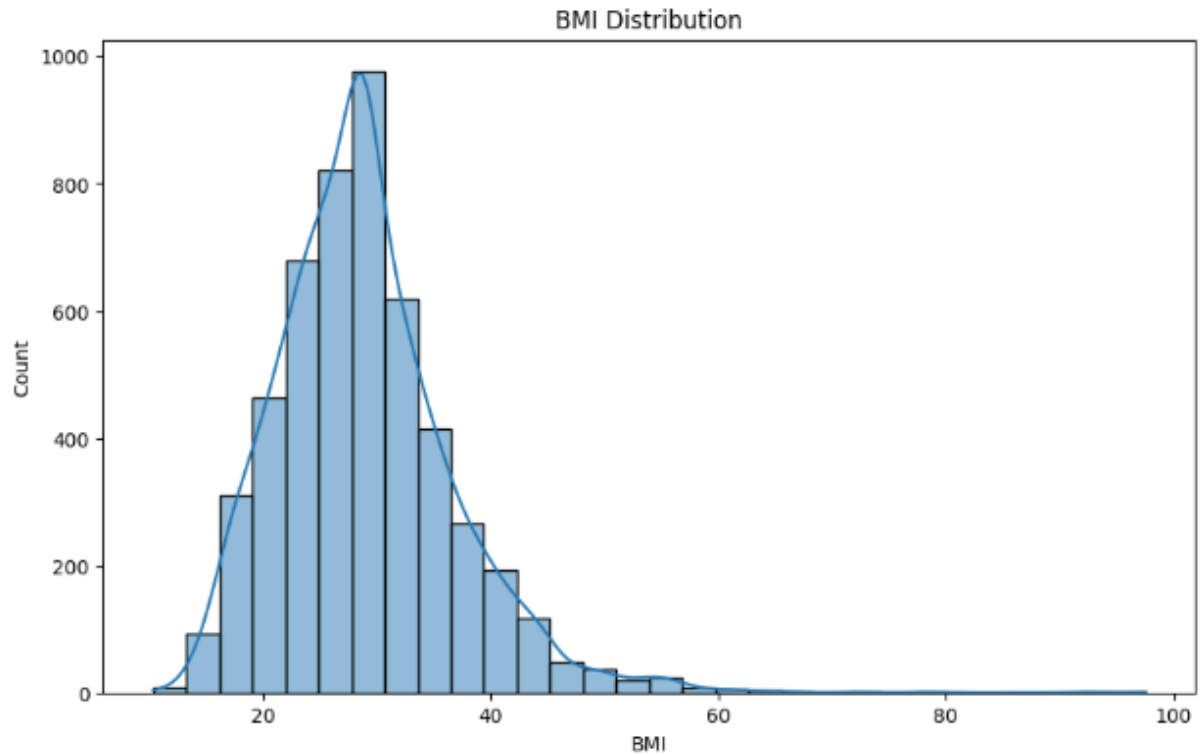
The correlation heatmap was also examined to identify relationships among numerical variables. Although Naïve Bayes assumes conditional independence among predictors, the heatmap provided useful insight into how variables such as age, hypertension, heart disease, glucose level, and BMI related to stroke outcomes.



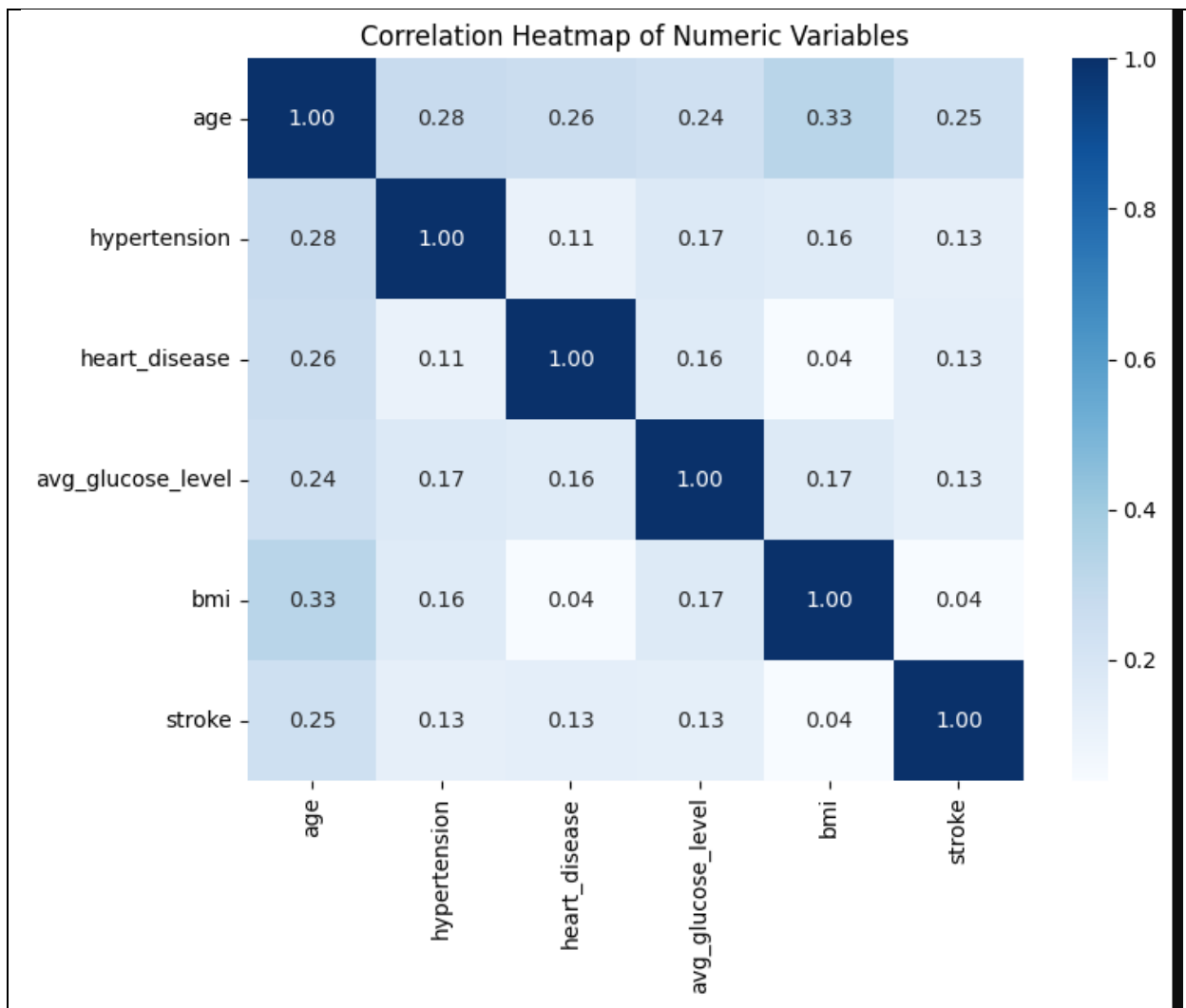
This figure illustrates the distribution of stroke and non-stroke cases within the dataset. The chart reveals a substantial class imbalance, with stroke cases representing only a small percentage of the total observations. This imbalance is important because it can significantly affect model training and evaluation, making metrics such as recall and ROC-AUC more informative than accuracy alone.



This histogram displays the distribution of patient ages within the dataset. The data include patients across a wide range of age groups. There is a concentration of observations among middle-aged and older adults. Since age is a well-established risk factor for stroke, this variable is expected to contribute meaningful predictive information to the classification model.



This figure shows the distribution of body mass index (BMI) values for patients in the dataset. Most observations are concentrated within the overweight BMI range, although several high-value outliers are present. BMI may contribute to stroke risk prediction because it is often associated with other health conditions such as hypertension and cardiovascular disease.



The correlation heatmap illustrates the relationships among the numerical variables in the dataset. Age, hypertension, heart disease, and average glucose level show positive relationships with stroke occurrence. Although Naïve Bayes assumes conditional independence among predictors, the heatmap provides useful insight into underlying data patterns before model fitting.

Preprocessing: armed with the exploratory analysis, perform the necessary preprocessing, both general and specific types appropriate for the modeling type being employed.

Several preprocessing steps were completed before model fitting. First, missing BMI values were replaced using the mean BMI value. After imputation there were no missing values remaining in the dataset.

The categorical variables gender, ever_married, work_type, Residence_type, and smoking_status were converted into numerical variables using one-hot encoding. This transformation was necessary because machine learning algorithms require numerical input variables.

The dependent variable was identified as stroke. All remaining predictive variables were used as independent variables. The id variable was removed because it serves only as an identifier and does not contain meaningful predictive information.

The dataset was divided into training and testing sets using a 75% training split and a 25% testing split. Stratification was applied during the split to preserve the proportion of stroke and non-stroke observations within both datasets. This was especially important because stroke cases were rare.

Unlike the Logistic Regression model used in Assignment 1, feature scaling was not required for Gaussian Naïve Bayes. The algorithm models class-specific probability distributions instead of optimizing coefficients through gradient-based techniques. As a result, the encoded dataset was used directly for model fitting.

Model Fitting: explain the key steps and activities you perform to fit the model. Experiment (as appropriate) with parameters tuning. This is key, what separates highly accurate model from a less accurate ones is the amount of performance tuning performed.

A Gaussian Naïve Bayes classifier was selected for this study because the dataset contains several continuous predictor variables. Gaussian Naïve Bayes estimates the probability distribution of each feature within each class. It applies Bayes' Theorem to calculate posterior probabilities.

One challenge encountered during model development was the severe class imbalance present in the dataset. Since only 4.9% of the observations represented stroke cases, the model risked favoring the majority non-stroke class. To address this issue, balanced class priors of [0.5, 0.5] were specified.

Parameter tuning was performed using the var_smoothing parameter. Multiple smoothing values ranging from 1e-12 to 1e-6 were evaluated using stratified cross-validation. Recall was

selected as the primary tuning metric. This was because identifying actual stroke cases was considered more important than maximizing overall accuracy.

The tuning process determined that a var_smoothing value of 1e-12 produced the highest mean recall score. The score was of approximately 0.995 during cross-validation. Therefore, the final Gaussian Naïve Bayes model was trained using balanced priors and the optimal smoothing parameter.

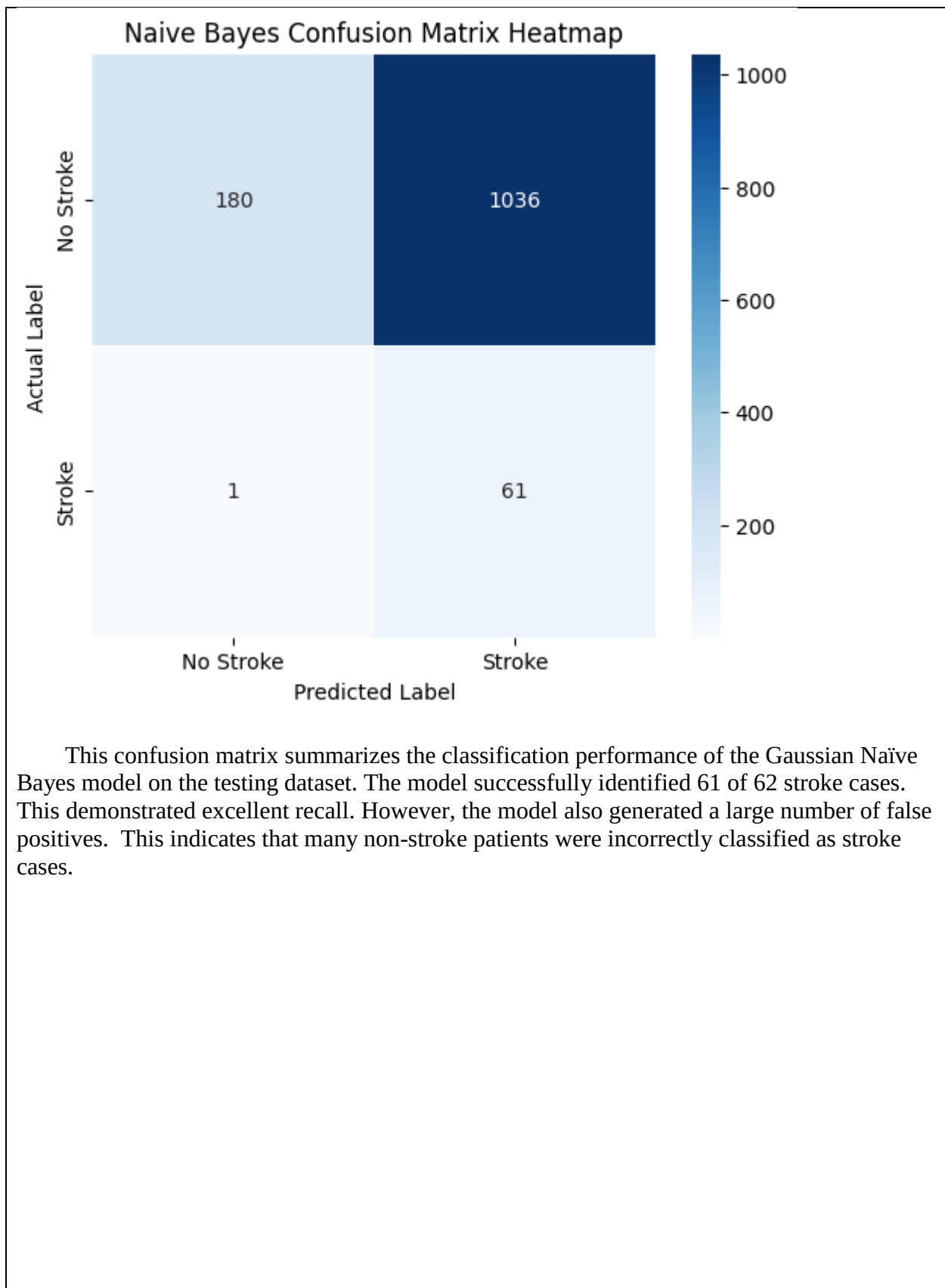
Results

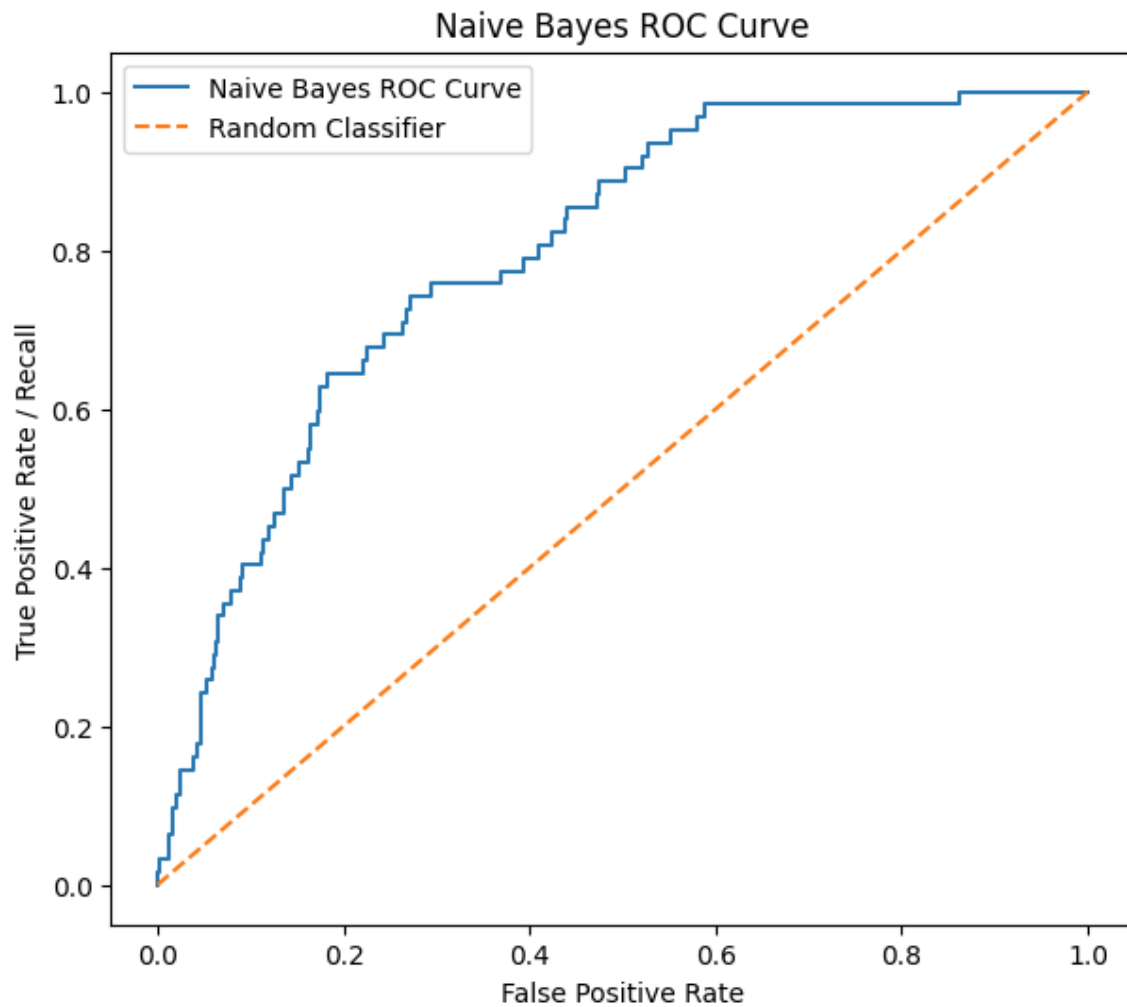
Model Properties: explain the components of the fitted model and their characteristics. Leverage functions to summarize the model properties. Also, leverage visualization as required.

The final Gaussian Naïve Bayes model used balanced class priors and a var_smoothing value of 1e-12. One advantage of Naïve Bayes is that it produces probabilistic predictions. It does not create simple class labels. These probabilities can assist with healthcare screening applications, especially where risk assessment may be more important than binary classification.

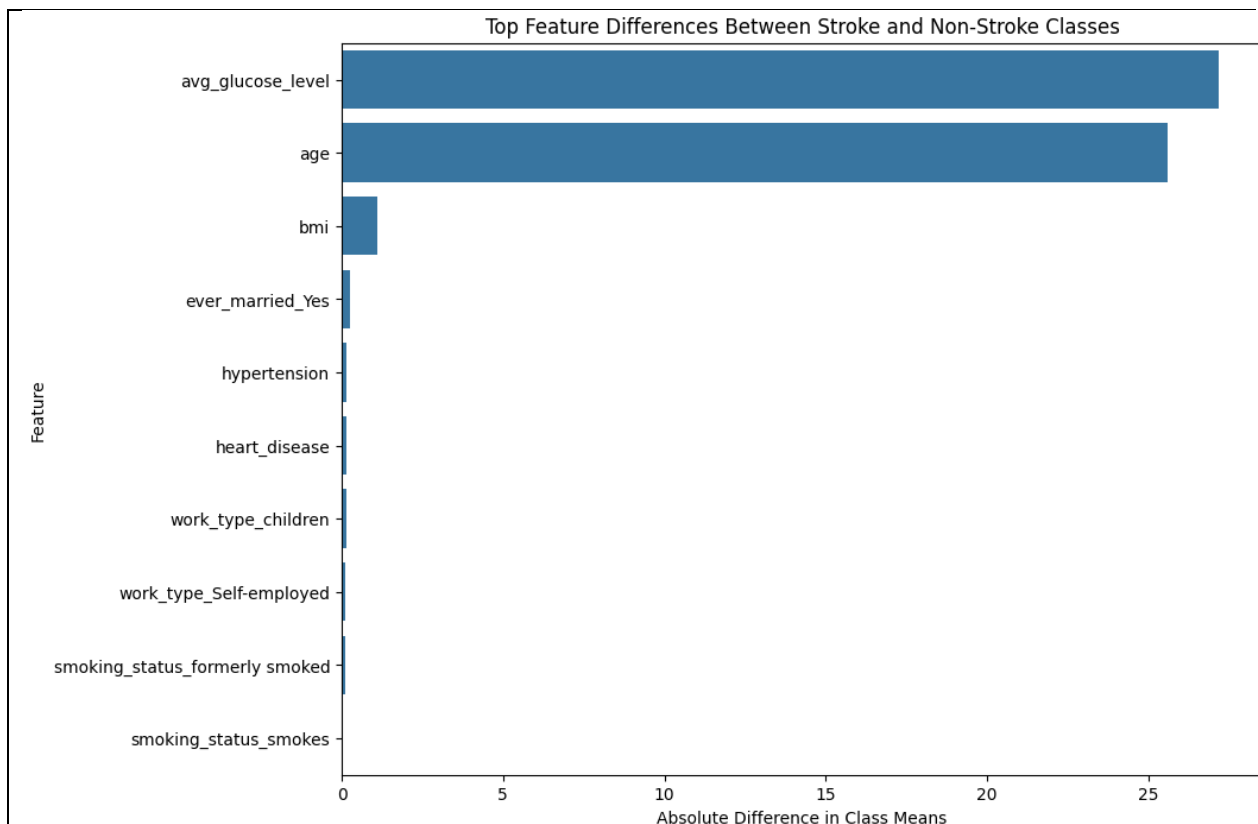
The model's class-conditional means provided insight into which variables differed most between stroke and non-stroke patients. Average glucose level exhibited the largest difference between classes. The stroke patient's glucose level averaged approximately 131.25 compared to 104.06 for non-stroke patients. Age also demonstrated a substantial difference. The stroke patients averaging approximately 67.64 years compared to 42.05 years for non-stroke patients.

Hypertension and heart disease were also more prevalent among stroke patients. These findings align with established medical knowledge regarding stroke risk factors. This suggests that the model successfully identified clinically meaningful patterns.





The ROC curve evaluates the model's ability to distinguish between stroke and non-stroke cases across multiple classification thresholds. The ROC-AUC score of approximately 0.796 indicates that the model performs substantially better than random guessing. This result suggests that the model learned meaningful predictive patterns despite its low overall classification accuracy.



This figure displays the variables with the largest differences in class-conditional means between stroke and non-stroke patients. Average glucose level and age exhibited the greatest differences. They were followed by BMI, hypertension, and heart disease. These findings suggest that the model identified clinically relevant characteristics associated with stroke risk.

Output Interpretation: explain the result and interpret the final model output using terms that reflect the application area and in relation to the stated objective. This is where you check whether or not the stated objective is met.

The confusion matrix revealed that the model correctly identified 61 of 62 stroke cases. It also misclassified only one actual stroke patient. This resulted in a recall score of approximately 0.98 for the stroke class. From a healthcare perspective, this represents a strong result because very few stroke cases were missed.

However, the model also incorrectly classified 1,036 non-stroke patients as stroke patients. This large number of false positives significantly reduced overall accuracy and

precision. The precision score for the stroke class was only 0.06. This indicates that many patients identified as stroke cases were actually non-stroke patients.

The overall accuracy score was approximately 18.86%. Although this value appears poor, it should be interpreted carefully due to the model's emphasis on stroke detection. The balanced priors encouraged the model to aggressively identify potential stroke cases rather than optimize overall classification accuracy.

The ROC-AUC score was approximately 0.796. This provides a more positive assessment of model performance. This value indicates that the model successfully learned meaningful relationships within the data. It was able to distinguish stroke patients from non-stroke patients substantially better than random guessing.

Personally, I found these results interesting. They demonstrate that a model can perform poorly on one metric while performing extremely well on another. The Naïve Bayes model prioritized sensitivity at the expense of precision and accuracy. This highlights the importance of selecting evaluation metrics that align with the problem being solved.

Evaluation: employ appropriate metrics to quantitatively evaluate the performance of the fitted model. For supervised classification, this includes simple accuracy, precision & recall (or sensitivity & specificity), all of which can be generated from a confusion matrix, or ROC.

The Gaussian Naïve Bayes model achieved an overall accuracy score of 18.86%. While this value is low, it does not fully reflect the model's effectiveness for stroke screening purposes. The severe class imbalance and use of balanced priors significantly influenced model behavior.

The model achieved a recall score of 0.98 for stroke cases. This means that approximately 98% of actual stroke patients were successfully identified. In healthcare applications, high recall is often desirable. This is because missing high-risk patients may lead to serious consequences.

The primary weakness of the model was its low precision score of 0.06 for the stroke class. Many non-stroke patients were incorrectly classified as stroke patients. This ended up producing a large number of false positives. This outcome would likely create additional costs and follow-up testing requirements if implemented in a real healthcare setting.

The ROC-AUC score of 0.796 suggests that the model learned meaningful patterns and possesses reasonable discriminatory ability. Although the classification threshold produced poor overall accuracy, the underlying probability estimates were substantially better than random guessing.

Overall, the Naïve Bayes model demonstrated excellent sensitivity but poor precision. This means it may be useful as an early screening tool but would not be appropriate as a standalone diagnostic system.

Conclusion

Summary: highlight the main findings in relation to the stated objective. You don't need to discuss the details of the analysis and the model such as accuracy here, just focus on the key findings.

This study applied Gaussian Naïve Bayes classification to predict stroke occurrence using demographic, lifestyle, and medical variables. The project included exploratory analysis, preprocessing, model fitting, output interpretation, and evaluation.

The results demonstrated that the model successfully identified nearly all stroke cases within the testing dataset. The model achieved a recall score of approximately 0.98. It correctly identified 61 of 62 stroke patients. This indicates that the model was highly effective at detecting patients who may be at risk.

However, this strong recall performance came at the expense of overall accuracy and precision. The model generated a large number of false positives. This resulted in an overall accuracy score of approximately 18.86%. Despite this limitation, the ROC-AUC score of 0.796 suggests that the model successfully learned meaningful relationships within the dataset.

One of the most important lessons I learned from this project is that machine learning performance cannot be evaluated using a single metric. Depending on the application, recall, precision, accuracy, or ROC-AUC may be more important. In this healthcare scenario, the ability to identify stroke cases was more valuable than maximizing overall accuracy.

Limitations & Improvement areas: discuss the limitations of the analysis and identify potential improvement areas for future work. This could be related to the data, algorithm, or a combination of the two.

There are several limitations that should be considered when interpreting the results. First, the dataset was highly imbalanced. The stroke cases represented less than 5% of observations. This imbalance significantly influenced model behavior and evaluation outcomes.

Second, Naïve Bayes assumes conditional independence among predictor variables. In reality, variables such as age, hypertension, heart disease, glucose levels, and BMI may be correlated. This assumption may limit the model's ability to capture complex relationships.

Third, the dataset lacked several potentially important stroke risk factors. These factors include family medical history, cholesterol levels, physical activity measures, and medication information. Additional variables may improve predictive performance.

Future improvements could include testing alternative machine learning algorithms. Other learning algorithms include applications such as Random Forest, Gradient Boosting, XGBoost, Support Vector Machines, and Neural Networks. Future studies could also investigate oversampling techniques such as SMOTE to address class imbalance while maintaining strong recall performance.

Overall, the Naïve Bayes model demonstrated the ability to identify stroke patients effectively but produced too many false positives to serve as a standalone diagnostic tool. Its greatest value appears to be as an early screening model that can support additional clinical evaluation.

Appendix

First Five Rows:

	id	gender	age	hypertension	heart_disease	ever_married	\
0	9046	Male	67.0	0	1	Yes	
1	51676	Female	61.0	0	0	Yes	
2	31112	Male	80.0	0	1	Yes	
3	60182	Female	49.0	0	0	Yes	
4	1665	Female	79.0	1	0	Yes	

	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	\
0	Private	Urban	228.69	36.6	formerly smoked	
1	Self-employed	Rural	202.21	NaN	never smoked	
2	Private	Rural	105.92	32.5	never smoked	
3	Private	Urban	171.23	34.4	smokes	
4	Self-employed	Rural	174.12	24.0	never smoked	

stroke

0	1
1	1
2	1
3	1
4	1

Dataset Info:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 5110 entries, 0 to 5109

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	id	5110 non-null	int64
1	gender	5110 non-null	object
2	age	5110 non-null	float64
3	hypertension	5110 non-null	int64
4	heart_disease	5110 non-null	int64
5	ever_married	5110 non-null	object
6	work_type	5110 non-null	object
7	Residence_type	5110 non-null	object
8	avg_glucose_level	5110 non-null	float64
9	bmi	4909 non-null	float64
10	smoking_status	5110 non-null	object
11	stroke	5110 non-null	int64

dtypes: float64(3), int64(4), object(5)

memory usage: 479.2+ KB

None

Missing Values Before Cleaning:

id	0
gender	0
age	0
hypertension	0
heart_disease	0
ever_married	0
work_type	0
Residence_type	0
avg_glucose_level	0
bmi	201


```

smoking_status      0
stroke              0
dtype: int64

Missing Values After Cleaning:
id                  0
gender              0
age                0
hypertension        0
heart_disease        0
ever_married        0
work_type           0
Residence_type      0
avg_glucose_level    0
bmi                 0
smoking_status      0
stroke              0
dtype: int64

Descriptive Statistics:

```

	id	age	hypertension	heart_disease \
count	5110.000000	5110.000000	5110.000000	5110.000000
mean	36517.829354	43.226614	0.097456	0.054012
std	21161.721625	22.612647	0.296607	0.226063
min	67.000000	0.080000	0.000000	0.000000
25%	17741.250000	25.000000	0.000000	0.000000
50%	36932.000000	45.000000	0.000000	0.000000
75%	54682.000000	61.000000	0.000000	0.000000
max	72940.000000	82.000000	1.000000	1.000000

	avg_glucose_level	bmi	stroke
count	5110.000000	5110.000000	5110.000000
mean	106.147677	28.893237	0.048728
std	45.283560	7.698018	0.215320
min	55.120000	10.300000	0.000000
25%	77.245000	23.800000	0.000000
50%	91.885000	28.400000	0.000000
75%	114.090000	32.800000	0.000000
max	271.740000	97.600000	1.000000


```

Encoded Dataset Preview:

```

	id	age	hypertension	heart_disease	avg_glucose_level	bmi	stroke
\							
0	9046	67	0	1	228	36	1
1	51676	61	0	0	202	28	1
2	31112	80	0	1	105	32	1
3	60182	49	0	0	171	34	1
4	1665	79	1	0	174	24	1

	gender_Male	gender_Other	ever_married_Yes	work_type_Never_worked	\
0	1	0	1		0
1	0	0	1		0
2	1	0	1		0

```

3          0          0          1          0
4          0          0          1          0

work_type_Private work_type_Self-employed work_type_children \
0          1          0          0
1          0          1          0
2          1          0          0
3          1          0          0
4          0          1          0

Residence_type_Urban smoking_status_formerly smoked \
0          1          1
1          0          0
2          0          0
3          1          0
4          0          0

smoking_status_never smoked smoking_status_smokes
0          0          0
1          1          0
2          1          0
3          0          1
4          1          0

Independent Variables:
Index(['age', 'hypertension', 'heart_disease', 'avg_glucose_level', 'bmi',
      'gender_Male', 'gender_Other', 'ever_married_Yes',
      'work_type_Never_worked', 'work_type_Private',
      'work_type_Self-employed', 'work_type_children',
      'Residence_type_Urban',
      'smoking_status_formerly smoked', 'smoking_status_never smoked',
      'smoking_status_smokes'],
      dtype='object')

Dependent Variable:
stroke

Naive Bayes Parameter Tuning Results:
var_smoothing mean_recall
0  1.000000e-12  0.994737
1  1.000000e-11  0.994737
2  1.000000e-10  0.994737
3  1.000000e-09  0.994737
4  1.000000e-08  0.978521
5  1.000000e-07  0.946230
6  1.000000e-06  0.913940

Best var_smoothing value based on recall:
1e-12

Confusion Matrix:
[[ 180 1036]
 [   1   61]]

Accuracy Score:
0.18857589984350548

```

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.15	0.26	1216
1	0.06	0.98	0.11	62
accuracy			0.19	1278
macro avg	0.53	0.57	0.18	1278
weighted avg	0.95	0.19	0.25	1278

ROC-AUC Score:

0.7956945033955857

Naive Bayes Class Conditional Means:

	age	hypertension	heart_disease	avg_glucose_level	\
No Stroke Mean	42.052126	0.088340	0.046365	104.062551	
Stroke Mean	67.636364	0.251337	0.187166	131.245989	

	bmi	gender_Male	gender_Other	ever_married_Yes	\
No Stroke Mean	28.370096	0.414266	0.000274	0.648560	
Stroke Mean	29.491979	0.422460	0.000000	0.893048	

	work_type_Never_worked	work_type_Private	\
No Stroke Mean	0.003567	0.568176	
Stroke Mean	0.000000	0.577540	

	work_type_Self-employed	work_type_children	\
No Stroke Mean	0.159396	0.141564	
Stroke Mean	0.272727	0.005348	

	Residence_type_Urban	smoking_status_formerly smoked	\
No Stroke Mean	0.504801	0.169822	
Stroke Mean	0.524064	0.272727	

	smoking_status_never smoked	smoking_status_smokes
No Stroke Mean	0.369273	0.149520
Stroke Mean	0.358289	0.176471

Top Feature Differences Between Stroke and Non-Stroke Classes:

	Feature	No_Stroke_Mean	Stroke_Mean	\
3	avg_glucose_level	104.062551	131.245989	
0	age	42.052126	67.636364	
4	bmi	28.370096	29.491979	
7	ever_married_Yes	0.648560	0.893048	
1	hypertension	0.088340	0.251337	
2	heart_disease	0.046365	0.187166	
11	work_type_children	0.141564	0.005348	
10	work_type_Self-employed	0.159396	0.272727	
13	smoking_status_formerly smoked	0.169822	0.272727	
15	smoking_status_smokes	0.149520	0.176471	
12	Residence_type_Urban	0.504801	0.524064	
14	smoking_status_never smoked	0.369273	0.358289	
9	work_type_Private	0.568176	0.577540	
5	gender_Male	0.414266	0.422460	

8	work_type_Never_worked	0.003567	0.000000
6	gender_Other	0.000274	0.000000
Absolute_Mean_Difference			
3	27.183438		
0	25.584237		
4	1.121883		
7	0.244488		
1	0.162997		
2	0.140801		
11	0.136216		
10	0.113331		
13	0.102906		
15	0.026951		
12	0.019263		
14	0.010984		
9	0.009365		
5	0.008194		
8	0.003567		
6	0.000274		

References

- Bati, F. (n.d.). *Lecture 3: Bayesian classification—Naïve Bayes* [PowerPoint slides]. University of Maryland Global Campus.
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